Bubbles and Priors: An AI-Agent Experiment

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Abstract. We replicate the finite-horizon experimental asset market of Dufwenberg et al. (2005) inside a deterministic browser-based simulator and extend it with the expected-utility messaging population of Lopez-Lira (2025). We then introduce *AI-Agent Prior Elicitation* (AIPE), a protocol in which each Lopez-Lira utility agent's starting subjective valuation is drawn, once per run, from a live AI-agent call whose prompt encodes the agent's risk preference, bias mode, and the presence of experienced counterparties. The returned psychological anchor $\hat{V}_i$ is clamped to $[0.25\,\mathrm{FV}_0,\ 1.75\,\mathrm{FV}_0]$, seeds the agent's prior on its first decision tick, and then yields to the standard Lopez-Lira belief-blending channel. The research question is whether the asset ultimately ends up in the hands of the agent with the highest anchored valuation. The testbed reports per-tick matching, a normalised valuation gap, Haessel R^2, normalised deviation, amplitude, turnover and allocative efficiency; the merged Experienced-plus-Utility population is split as $\lceil (N - N_{\text{fixed}})/2 \rceil$ experienced and the remainder utility. The simulator preserves reproducibility through a seeded mulberry32 PRNG and a deterministic fallback when the AI-agent call fails, so every finding can be replayed without network access. This paper documents the replication, the new anchor channel, and the market-quality instruments used to answer the allocation question.

Keywords: experimental asset markets, bubbles, expected utility, AI agents, prior elicitation, allocative efficiency.

JEL Codes: C92, D84, G12, G41.

I. Introduction

Experimental asset markets in the tradition of Smith et al. (1988) have produced one of the most replicated departures from rational-expectations pricing in all of economics: even when subjects know, in closed form, that the fundamental value of a finite-lived asset declines deterministically with the remaining dividend stream, transaction prices systematically overshoot the fundamental before collapsing near the terminal period. Dufwenberg et al. (2005)—hereafter DLM 2005—established a by now canonical result in this literature: when two-thirds or more of the subject pool consists of traders who have played the game before, the bubble disappears. Experience, rather than cognition or incentives, is what eventually kills the bubble.

A parallel and much more recent literature replaces the human subjects with AI agents. Lopez-Lira (2025) proposes an expected-utility framework in which each AI-agent trader carries a persistent valuation bias b_i, a risk preference over wealth $U \in \{U_L, U_N, U_A\}$, and a belief-blending channel that lets agents exchange private valuations and update a trust score for each sender by exponential moving average. Lopez-Lira's agents optimise expected utility over a small action set—hold, cross the bid, cross the ask, post passively—and the resulting population can be dialled between skeptical and naive, truthful and deceptive, risk-averse and risk-loving by simple categorical knobs. The Lopez-Lira design is a natural vehicle for studying how a heterogeneous AI-agent population prices a DLM asset, but it leaves open a psychological question the original DLM protocol never asked: if one replaces the deterministic prior $\mathrm{FV}_t \cdot (1 + b_i + \varepsilon)$ with a private

*Computational replication and source code are available at `https://github.com/Mon-ius/bubble`. The author thanks the maintainers of the original DLM and Lopez-Lira testbeds for making their designs easy to re-implement. All errors are the author's own.

subjective valuation elicited from an AI agent that is told who the trader is and what the market looks like, does the asset then migrate, over the course of the run, into the hands of the agent with the highest such valuation?

This paper contributes a computational testbed in which the two designs share one continuous double-auction market, and a new protocol—*AI-Agent Prior Elicitation* (AIPE)—that seeds each utility agent's prior with exactly one AI-agent call at run start. Concretely, when AIPE is enabled and the user supplies an API key, the simulator fires one chat completion per utility agent in parallel, using a prompt that encodes the agent's risk preference, bias mode, and the presence of DLM experienced counterparties in the same market. The returned number, clamped to $[0.25\,\mathrm{FV}_0,\ 1.75\,\mathrm{FV}_0]$, is written to the agent's `psychAnchor` slot and consumed on the first decision tick in place of the deterministic prior; subsequent ticks revert to the Lopez-Lira prior plus belief blending. In the notation of the replicated model,

$$\hat{V}_i \ \leftarrow\ \mathrm{AIPE}(\mathrm{agent}_i), \qquad \mathrm{prior}_{t=0}^{(i)} \ =\ \hat{V}_i \cdot (1+\varepsilon_i), \qquad \hat{V}_i \in [0.25\,\mathrm{FV}_0,\ 1.75\,\mathrm{FV}_0]. \tag{1}$$

The deterministic fallback—used when no key is supplied or the network call fails—drops equation (1) and resumes the Lopez-Lira prior $\mathrm{FV}_t \cdot (1 + b_i + \varepsilon)$ unchanged. Because the run then proceeds through a deterministic seeded PRNG, every simulation can be replayed identically with or without the anchor channel.

The research question is easy to instrument. Let $i^\star = \arg\max_{i \in U} \hat{V}_i$ be the utility agent with the highest anchored valuation and $j^\star(t) = \arg\max_i \mathrm{inventory}_i(t)$ the top holder at tick t; a *match* at t is $\mathbb{1}\{j^\star(t) = i^\star\}$ and the *normalised gap* is $g(t) = (\hat{V}_{i^\star} - \hat{V}_{j^\star(t)})/\hat{V}_{i^\star}$. Both quantities are reported every tick alongside Haessel R^2, normalised deviation, amplitude, turnover, and allocative efficiency, so classical bubble statistics stay visible beside the new allocation metrics.

Two features distinguish AIPE from a naïve "AI-as-trader" pipeline. First, the AI agent is consulted exactly once at run start; after the anchor is absorbed, the trader is again a pure expected-utility maximiser and the mulberry32 RNG alone controls the downstream dynamics. Second, the anchor is a *valuation* and not an order: the trader's quoting rule, fill probability, and utility function are unchanged, and only its starting prior is moved. These two choices isolate the specific question "does psychology predict ownership" from the adjacent question "does an AI agent trade well."

Section II places the simulator in the literature on experimental asset markets and AI-agent traders. Section III documents the market structure and the sampling stage. Section IV gives the precise decision rule for each agent type. Section V presents AIPE: the elicitation call, the clamp, the consumption path, and the research-question instrumentation. Section VI reports preliminary findings, and Section VII concludes.

II. Related Literature

The simulator sits at the intersection of four strands of literature: the experimental-bubble tradition, the heterogeneous agent-based finance tradition, the expected-utility decision-theoretic tradition, and the emerging work on AI agents as economic subjects.

Experimental bubbles.. The taxonomy of the market in this paper is directly inherited from Smith et al. (1988), who introduced the declining-fundamental, finite-horizon asset and documented the characteristic bubble-and-crash trajectory in a continuous double auction. Dufwenberg et al. (2005) show that the Smith–Suchanek–Williams bubble is fragile to *experience* rather than to instructions, incentives, or trading institutions: once two-thirds of the subjects have played the game before, transaction prices track the fundamental staircase from the first period onward. This result motivates the horizon-discount rule used by the simulator's *Experienced* (E) agent, in which an experience level $\varphi(h)$ ramps belief from FV_t in early rounds to zero in the last three periods.

Heterogeneous agent-based finance.. Three classical rules supply the pinned background of the simulator: a *Fundamentalist* (F) that mean-reverts toward FV_t in the Smith–Suchanek–Williams lineage; a *Trend follower* (T) that implements the positive-feedback strategy of De Long et al. (1990) by extrapolating a six-tick slope; and a *Random* (R) zero-intelligence trader after Gode and Sunder (1993) that posts uniform bids on $[0.7, 1.1]\,\mathrm{FV}_t$ and uniform asks on $[0.9, 1.3]\,\mathrm{FV}_t$. Together these three rules provide a rational anchor, a momentum amplifier, and a liquidity floor; the DLM experienced agent then contributes the horizon-conditional

bubble-suppression channel.

Expected utility and AI-agent traders.. Lopez-Lira (2025) introduces a utility-maximising AI-agent trader whose prior is a perturbation of FV_t, whose action space is $\{\text{hold, cross bid, cross ask, passive bid, passive ask}\}$, and whose choice rule picks the action with the highest expected utility, $\mathrm{EU}(a) = p_{\text{fill}} \cdot U(w_1) + (1 - p_{\text{fill}}) \cdot U(w_0)$. Agents carry one of three risk-preference functionals $\{U_L(w), U_N(w), U_A(w)\} = \{(w/w_0)^2,\ w/w_0,\ \sqrt{w/w_0}\}$ and exchange private valuations through a lightweight messaging bus. A pairwise trust score $\tau_{r\to s} \in [0, 1]$ is updated by exponential moving average whenever a claimed valuation is observed against the period's volume-weighted average trade price. The present paper inherits this framework without modification and reimplements it exactly as described, so that any difference from a pure Lopez-Lira run is attributable to the ChatGPT-anchored prior rather than to a change in the decision rule.

AI agents as economic subjects.. A small but rapidly growing literature studies whether AI agents reproduce well-documented behavioural regularities when embedded in experimental decision tasks. AIPE is a deliberately narrow use of this approach. It does not ask the AI agent to trade; it does not expose the order book or the history of transaction prices; it does not grant the AI agent the authority to choose an action from the action set. It asks one question—"what is your private subjective valuation per share right now, in cents?"—with a prompt that names the trader's risk preference, bias mode, and the composition of the market. The returned number is then clamped and treated as a prior in the downstream expected-utility computation. This design isolates the psychological channel ("how would the AI agent price the asset, in light of who it is?") from the trading channel ("what orders would the AI agent place?"), and in doing so connects the experimental bubble literature to the psychology-of-valuation literature via a single numeric quantity.

Market-quality instruments.. The testbed reports four standard bubble statistics—Haessel's R^2 (Haessel, 1978), normalised absolute deviation, amplitude, and turnover, following SSW and Stöckl et al. (2010)—alongside allocative efficiency $\mathrm{AE} = (\sum_i \hat{V}_i \cdot \text{inventory}_i)/(\max_i \hat{V}_i \cdot Q)$, the quantitative form of the AIPE research question: when $\mathrm{AE} = 1$ the asset has migrated to the agent with the highest anchored valuation.

III. Experimental Design

A. Session structure: four consecutive rounds

The unit of study in Dufwenberg et al. (2005) is a *session* of four consecutive ten-period markets played by the same six traders. The paper insists on the terminology explicitly:

> "A session involved four consecutive markets. In the following, we shall talk in terms of four different rounds. Note the distinction between rounds and periods; a round (being a market) consists of ten periods." (Dufwenberg et al., 2005, p. 1733)

The session wrapper is the load-bearing element of the original design: bubbles inflate in round 1, attenuate in rounds 2–3, and have effectively died by round 4 even though the population is identical—experience, not composition, kills the bubble. The simulator therefore wraps the per-round market in a session loop of $R = 4$ rounds. At every round boundary the order book is cleared, each agent's cash and inventory are rewound to its sampled spec, and the period counter resets to one. Learned state—trust matrices, belief modes, the EU candidate cache—is intentionally preserved across the boundary so that the cross-round learning channel that drives the paper's headline result is the only source of persistent change between consecutive markets. Both R and T are surfaced read-only in the simulator's "Paper constants" panel and are not user tunable.

B. The asset and the fundamental

The asset is a finite-lived claim on an iid per-period dividend that pays either zero or twenty experimental cents with equal probability. Letting d denote the period-t dividend draw,

$$d \sim \text{Uniform}\{0, 20\}, \qquad \mathbb{E}[d] = 10, \qquad T = 10. \tag{2}$$

The asset lives for $T = 10$ periods within each round and pays nothing thereafter, so its risk-neutral fundamental value at the start of period t of any round is

$$\mathrm{FV}_t \;=\; \mathbb{E}[d] \cdot (T - t + 1) \;=\; 10 \cdot (T - t + 1), \tag{3}$$

that is, a linearly declining staircase from $\mathrm{FV}_1 = 100$ to $\mathrm{FV}_T = 10$ with a terminal value of zero after the final dividend pays. The fundamental resets to $\mathrm{FV}_1 = 100$ at the start of every new round, so the price chart traces a four-tooth saw-tooth across the full session. Equation (2) and equation (3) reproduce the dividend process and the fundamental schedule of Dufwenberg et al. (2005) exactly. The session length R, the per-round asset life T, and the dividend mean $\mathbb{E}[d]$ are all surfaced read-only in the simulator's "Paper constants" panel so that every comparison across populations holds the market structure fixed.

C. Continuous double auction with period discretisation

The simulator discretises each period into $K = 18$ decision rounds, so a single round comprises $T \cdot K = 180$ ticks and a full four-round session comprises $R \cdot T \cdot K = 720$ ticks. Every tick, each agent posts at most one order, routed to a price–time priority order book; a crossing pair matches at the resting price. At the end of each period the outstanding quotes are withdrawn and the dividend is drawn from (2) and credited to each holder. At the end of period T of every non-final round the session loop also rewinds endowments, clears the book, and resets the period counter to one before the next round begins. The $K = 18$ discretisation is the simulator's own approximation of the two-minute continuous z-Tree window used in the paper, is pinned in the "Simulator constants" panel, and is not user tunable.

D. Strict-DLM mode: fixed $N = 6$ and two discrete endowments

The simulator's *strict-DLM mode* replicates the original protocol without relaxation. Selecting the DLM 2005 paradigm in the navbar pins $N = 6$, disables the population sliders, and routes the sampling stage through a dedicated two-type draw. The paper specifies exactly six traders per session:

> "At each session, six subjects participated in a sequence of four consecutive markets for an experimental asset." (Dufwenberg et al., 2005, p. 1733)

and exactly two discrete endowment types with identical implied buy-and-hold value:

> "Before a market opened, half of the traders started with 200 cents and six assets, while each of the other traders started with 600 cents and two assets." (Dufwenberg et al., 2005, p. 1733)

Both bundles have buy-and-hold value $E[\text{cash}] = 200 + 6 \cdot \mathbb{E}[d] \cdot T = 200 + 600 = 800$ for type A and $600+2\cdot\mathbb{E}[d]\cdot T = 600+200 = 800$ for type B, so the split is strictly distributional: it varies the cash–inventory mix across subjects without creating value heterogeneity.[1]

The simulator implements this with `dlmSampleAgents(rng)`, which pins six slots, assigns the index sequence $[0, 0, 0, 1, 1, 1]$ (three type A, three type B), Fisher–Yates shuffles it against the dedicated sampling RNG, and writes the matching cash/inventory pair onto each spec. No other mode touches the two-type draw: Lopez-Lira and AIPE continue to use the relaxed $U[800, 1200] \times \{2, 3, 4\}$ sampler described for backwards compatibility with their respective strategy-cube populations.

E. Endogenous experience

The original study is unusual in experimental economics in that it treats experience as a *procedural* rather than compositional variable: subjects do not differ by type at the start of a session, and no one is ever recruited because they are "experienced". Experience is a property of the session loop: a subject acquires it by playing rounds 1, 2, and 3 alongside five peers in a shared six-person market, and the paper's central finding is that this procedural experience — not a rule-change or a population swap — is what kills the bubble by round 4.

Reproducing DLM's design therefore requires that experience not exist as a pre-assignable agent *type*. The simulator uses a single class, `DLMTrader`, with a two-state `decide` method gated on an integer counter `roundsPlayed` that the engine increments at every round boundary:

$$\texttt{DLMTrader}.\texttt{decide}(\cdot) = \begin{cases} \texttt{inexperienced}(\cdot), & \text{if } \texttt{roundsPlayed} = 0, \\ \texttt{experienced}(\cdot), & \text{if } \texttt{roundsPlayed} \geq 1. \end{cases} \tag{4}$$

[1] More directly, each bundle has the same risk-neutral fundamental value $\mathrm{FV}_1 = 100$ per share at round start, so 200+6 · 100 = 600+2 · 100 = 800 + 200 = 1,000 ¢ once the initial cash is counted.

Every trader is instantiated with `roundsPlayed` $= 0$, so at the start of round 1 every participant is operating on the inexperienced branch: it chases trend, accepts zero-intelligence noise, ignores the fundamental horizon, and does not liquidate late in a round. The engine's round-end hook then increments the counter for every agent that has just finished a round, so the same six traders enter round 2 having flipped to the experienced branch, which anchors bids to $\varphi(h) \cdot \mathrm{FV}_t$ with horizon $h = T - t + 1$, refuses to buy above belief in late periods, and liquidates its remaining inventory in the final three periods of a round. Nothing else changes about the session; the same numeric id, cash, inventory and trust state persist — only the decision branch flips.

Crucially, no agent is ever instantiated with a positive `roundsPlayed`. The bubble-killing channel is therefore purely procedural, matching Dufwenberg et al. (2005) exactly.

F. Round-4 replacement and treatments

The paper introduces two treatment conditions that manipulate the mix of experienced and inexperienced traders at the start of round 4:

> "In the fourth round, depending on treatment, two or four experienced subjects who had participated in the first three rounds were randomly selected, removed, and replaced by the same number of inexperienced subjects." (Dufwenberg et al., 2005, p. 1733)

The simulator fires the replacement between rounds 3 and 4 whenever strict-DLM mode is active. Let $k \in \{2, 4\}$ denote the treatment size (T2 or T4). The engine's round-end branch runs, in order:

1. snapshot each surviving agent's period-T cash into `roundFinalCash`$[r-1]$ (Section H),
2. log a `round_end` event,
3. increment every surviving agent's `roundsPlayed` by one, so the four or two survivors of the T2 (resp. T4) sweep carry `roundsPlayed`=3 into round 4,
4. Fisher–Yates select k agents uniformly at random from the pool of surviving non-replacement traders and remove them,
5. draw k fresh `DLMTrader` specs from `dlmSampleReplacementAgent` (each with a new name drawn from the remaining name pool so that no two seats share a display name, and a fresh endowment type A or B drawn iid) and instantiate them with `roundsPlayed` $= 0$ and a `replacementFresh` flag for UI badging,
6. splice the new instances back into the engine at the vacated numeric ids, so the order book, logger, and replay system continue to key by a stable id,
7. rewind every surviving agent's cash and inventory to its spec (the fresh agents take their own replacement endowment on the same call), increment the round counter to 4, reset `period=1`, and clear the book.

The net effect is that round 4 is played with $6 - k$ veterans on the experienced branch of equation (4) and k fresh subjects on the inexperienced branch — precisely the composition DLM's treatment manipulates.

G. Multi-session batch

Dufwenberg et al. (2005) report ten sessions in total, five per treatment, interleaved so that the overall design has $10 \times 4 = 40$ markets. The simulator provides a one-click batch runner that reproduces this: pressing "Run 10-session batch" in the DLM panel drives ten synchronous `Engine.runToEnd()` loops in two blocks of five, alternating treatment, with an independent engine seed per session and an independent two-type endowment draw. Each session writes a compact summary row — treatment, mean $|p_t - \mathrm{FV}_t|/\mathrm{FV}_t$, per-share turnover, total trade count, mean per-subject payoff, and the round-4 replacement record — into a results table in the DLM panel. The batch preserves the pre-batch interactive state so the user's current session is not disturbed.

H. Payoff accounting

Session payoffs in DLM 2005 have two components: the subject's final cash at the end of each round, summed across all four rounds, plus a show-up fee of $5:

> "Subjects were privately paid, in cash, the amount of their final cash holdings from each round. They were also paid a show-up fee of $5." (Dufwenberg et al., 2005, p. 1735)

The simulator captures each surviving agent's cash at the end of period T of every round via `Logger.logRoundFinalCas` which writes a per-agent `{id: cash}` map into `roundFinalCash`$[r-1]$. For agent i the session payoff is therefore

$$\pi_i \;=\; \sum_{r=1}^{R} \texttt{roundFinalCash}[r-1][i] \;+\; 500, \tag{5}$$

where the 500 cents show-up fee is added unconditionally and shares held at the end of a round are worth nothing because the asset's life has ended (the dividend stream has paid out). Agents spliced in by the round-4 replacement step only contribute a round-4 row to the sum because they did not play rounds 1–3; this matches the paper's accounting for replacement subjects, who are paid only for the single round they participate in plus the show-up fee.

I. Departures from the original protocol

The simulator reproduces DLM 2005 without exception except in one place: the original study runs a continuous two-minute open-outcry double auction in z-Tree for each period (Dufwenberg et al., 2005, p. 1733), whereas the simulator discretises that window into $K = 18$ decision rounds in order to run a deterministic tick-level loop that the replay system can slice. Eighteen is dense enough to reproduce the bubble–crash pattern at qualitative parity with the paper while keeping the replay buffer compact; lower values starve the book, higher values compress the per-tick update so agents barely respond between decisions. This is the *only* residual simulator approximation. No other relaxation remains: the population size, the two endowment types, the round count, the dividend distribution, the fundamental schedule, the endogenous experience gate, the round-4 treatment, the ten-session batch, and the payoff + show-up fee structure are all in strict agreement with the paper.

The Lopez-Lira and AIPE paradigms retain the $N \geq 4$ relaxed sampler and the mixed F/T/R background described elsewhere in this paper, since they are *extensions* rather than replications of DLM and therefore intentionally depart from the strict design at the populations layer. Strict-DLM mode is the single-button knob that turns every one of those relaxations off.

J. Reproducibility

Every run is driven by a single scalar seed. A mulberry32 generator $s_{k+1} = (s_k + \texttt{0x6D2B79F5}) \bmod 2^{32}$ produces the $(0,1)$-uniform stream consumed by every stochastic step—dividend realisations, zero-intelligence quotes, the noise term in the utility prior, and the trust-update ordering. Calls to `Math.random` from agent or market code are a contract violation that breaks reproducibility by the (population, seed) pair and are flagged as an invariant in the project style guide. The seed is re-rolled only on Reset, and the replay system reconstructs past states by slicing append-only history arrays on `Market` and `Logger`—no frame is ever recomputed. A single seed therefore reproduces the entire 720-tick session end-to-end, rounds and round-end resets included.

IV. Agent Model

This section gives the exact decision rule for each of the five agent classes in the simulator. All rules read the top of the book (best bid b_t and best ask a_t), the current fundamental value FV_t, and a seeded pseudo-random draw $r \sim \mathrm{Uniform}(0,1)$; the formulas below are the literal rules implemented in the source code, with constants exposed to the reader so that any experiment can be reconstructed from this page alone. The *DLM trader* class (§D) has two internal decision branches selected by the endogenous `roundsPlayed` counter described in §E; every other class has a single branch.

A. Fundamentalist (F)

The fundamentalist is a rational mean-reversion anchor inspired by Smith et al. (1988). Its belief is $\beta_t = \mathrm{FV}_t \cdot \big(1 + 0.02\,(r - 0.5) \cdot 2\big)$—a $\pm 2\%$ symmetric noise about the fundamental—and its action rule is

$$\begin{cases} \text{cross the ask with a unit bid at } a_t, & \text{if } a_t < 0.98\,\beta_t \text{ and cash} \geq a_t, \\ \text{cross the bid with a unit ask at } b_t, & \text{if } b_t > 1.02\,\beta_t \text{ and inventory} > 0, \\ \text{post passive with probability } 0.45, & \text{otherwise}, \\ \text{hold}, & \text{otherwise}. \end{cases} \tag{6}$$

Passive quotes bid in $[0.970,\ 1.005] \cdot \beta_t$ and ask in $[0.995,\ 1.030] \cdot \beta_t$, so the fundamentalist never posts a cross on its own initiative. It supplies a narrow two-sided book near the fundamental whenever neither side of the market is aggressive and anchors mispricing back to FV_t through unit crosses when either side drifts by more than 2%.

B. Trend follower (T)

The trend follower estimates a short-horizon slope from the last six non-null trade prices and projects three ticks forward,

$$\text{slope}_t \;=\; \frac{p_t - p_{t-h+1}}{\max(1, h-1)}, \qquad \text{target}_t \;=\; p_t + 3 \cdot \text{slope}_t, \tag{7}$$

where $h = 6$ is the slope window and p_t is the most recent trade price. When $\text{slope}_t > 0$ and it has cash, the trend agent chases up with a unit bid at $p_t \cdot (1 + 0.015 + 0.03\,r)$; when $\text{slope}_t < 0$ and it has inventory, it flees with a unit ask at $p_t \cdot (1 - 0.015 - 0.03\,r)$. Two corner cases protect the agent from stranding. First, if cash has dropped below the most recent trade price but inventory still exceeds the initial endowment, the agent dumps shares into the best bid at the aggressive price. Second, in a flat-slope regime with excess inventory, the agent liquidates with probability $\min(0.85,\ 0.3 + 0.25\,e)$, where e is the excess-inventory rate. The trend rule is the positive-feedback channel (De Long et al., 1990) and is the primary amplifier of bubble formation in the simulator.

C. Random zero-intelligence (R)

The zero-intelligence trader posts uniform bids and asks on fundamentally-scaled support, following Gode and Sunder (1993). Explicitly,

$$\text{bid}_t \sim \text{Uniform}[\,0.7\,\mathrm{FV}_t,\ 1.1\,\mathrm{FV}_t\,], \qquad \text{ask}_t \sim \text{Uniform}[\,0.9\,\mathrm{FV}_t,\ 1.3\,\mathrm{FV}_t\,]. \tag{8}$$

It never reads the order book, never tracks profit, and is pinned in every run so that the book never goes fully empty. One slot of this type is part of the fixed background and is invisible to the population-composition sliders.

D. DLM trader (D): endogenous experience

The DLM trader is the *only* agent class instantiated in strict-DLM mode and it implements the paper's endogenous-experience design exactly: there are no "naive", "once-played", or "veteran" types to pick between at sampling time; there is one class whose `decide` method reads `roundsPlayed` $\in \{0, 1, 2, \ldots\}$ and branches.

Inexperienced branch (`roundsPlayed` $= 0$).. A newly-instantiated DLM trader has zero information about the fundamental horizon and no liquidation urge in the terminal periods of a round. The behaviour is deliberately bubble-prone: trend-chasing on the most recent trade price with zero-intelligence noise layered on top. Let p_t be the most recent trade price (with FV_1 used when the book is empty). On each tick the agent flips between a unit bid and a unit ask with equal probability; the bid is drawn from $\text{Uniform}[0.9\,p_t,\ 1.15\,p_t]$ and the ask from $\text{Uniform}[0.9\,p_t,\ 1.2\,p_t]$, clipped by cash and inventory respectively. It never refuses to overpay, never liquidates late in a round, and in particular never consults FV_t.

Experienced branch (`roundsPlayed` ≥ 1).. Once the engine has incremented the counter (which happens exactly at round boundaries in the session loop), the same instance switches to an FV-anchored rule. Let

$h = T - t + 1$ be the number of periods remaining, and let $\varphi(h) = \varphi_0 + (h/T) \cdot \varphi_1$ be the horizon discount with $\varphi_0 = 0.92$ and $\varphi_1 = 0.06$ (the veteran calibration in DLM's terminology). The belief is

$$\beta_t \;=\; \mathrm{FV}_t \cdot \varphi(h). \tag{9}$$

When $h \leq 3$ the agent liquidates any remaining inventory into the best bid; when $h \leq 4$ it refuses to cross any ask priced above $0.9\,\beta_t$. Outside the terminal region it behaves like a risk-tight fundamentalist: it crosses the ask with a unit bid at a_t when $a_t < 0.92\,\beta_t$ and crosses the bid with a unit ask at b_t when $b_t > 1.05\,\beta_t$, otherwise posting a passive quote near β_t.

Instantiation.. Every DLM trader is born with `roundsPlayed` $= 0$. There is no pre-assignment knob, no sampling distribution over experience, no "round at which this agent became experienced" field. The only way an agent ever reaches the experienced branch is by surviving a round-end transition in a live session, which is why the round-4 replacement step (Section F) works: the fresh traders spliced in at $r = 4$ reset to the inexperienced branch while the survivors retain their branch flip from earlier in the session.

E. Utility (U)

The utility agent follows Lopez-Lira (2025). Its prior at every decision tick is

$$V_i^{\text{prior}} \;=\; \mathrm{FV}_t \cdot \big(1 + b_i + \varepsilon_{i,t}\big), \qquad b_i \in \{-0.15,\, 0,\, +0.15\}, \qquad \varepsilon_{i,t} \sim \text{Uniform}[-0.03,\, 0.03]. \tag{10}$$

The persistent bias b_i is the "persistently optimistic/pessimistic/unbiased" archetype and is assigned at sampling time. The per-tick noise ε keeps the agent's quoting out of lockstep with other agents of the same type.

The prior is then blended with incoming messages from peers. If $\mathcal{M}_i$ denotes the set of claimed valuations received during the current period, and $w \in (0, 1)$ is a weight, the posterior is

$$V_i^{\text{post}} \;=\; w \cdot V_i^{\text{prior}} \;+\; (1 - w) \cdot \frac{1}{|\mathcal{M}_i|} \sum_{m \in \mathcal{M}_i} \hat{v}_m. \tag{11}$$

Three archetypes determine w: *naive* trusts peers and uses $w = 0.60$; *skeptical* barely moves and uses $w = 0.90$; *adaptive* scales w by per-sender trust and caps the move at 0.50, with pairwise trust updated every trade by

$$\tau_{r \to s} \;\leftarrow\; (1 - \alpha)\, \tau_{r \to s} \;+\; \alpha \cdot \max\!\Big(0,\, 1 - \tfrac{|\hat{v}_s - \mathrm{VWAP}_t|}{\mathrm{VWAP}_t}\Big), \qquad \alpha = 0.30. \tag{12}$$

Receiver r's trust in sender s is reinforced whenever s's claim is close to the period VWAP and attenuated otherwise.

Given the posterior valuation V_i^{post}, the agent scores each available action $a \in \{\text{hold, cross bid, cross ask, passive bid, pass}$ by its expected utility

$$\mathrm{EU}(a) \;=\; p_{\text{fill}} \cdot U(w_1) \;+\; (1 - p_{\text{fill}}) \cdot U(w_0), \qquad a^\star = \arg\max_a \mathrm{EU}(a), \tag{13}$$

where w_0 and w_1 are the agent's wealth under non-fill and fill respectively, and $p_{\text{fill}} = 0.30$ is a constant passive fill-probability inherited from the original Lopez-Lira design. The risk-preference functional $U \in \{U_L, U_N, U_A\}$ is assigned at sampling time and drawn from

$$U_L(w) = (w/w_0)^2, \qquad U_N(w) = w/w_0, \qquad U_A(w) = \sqrt{w/w_0}. \tag{14}$$

A deceptive variant of the utility rule is available for the messaging channel. When enabled, an asset-poor agent (inventory below its initial endowment) broadcasts a valuation equal to $0.82 \cdot V_i^{\text{true}}$, while an asset-rich agent broadcasts $1.18 \cdot V_i^{\text{true}}$. The lie-gap magnitude $|\hat{v}^{\text{report}} - V^{\text{true}}|$ drives the period-end trust update in (12), which produces the closed-loop feedback described in Lopez-Lira (2025).

Equations (6)–(14) constitute the complete decision model. The market aggregates these orders through a price–time-priority order book, mutates cash and inventory on every trade, credits dividends per (2) at period end, and records every state in append-only history arrays so the replay system can reconstruct past ticks without re-running the agents. No trader in this model consults an AI agent at decision time. The next section adds the first such consultation as the *AI-Agent Prior Elicitation* (AIPE) protocol.

V. AI-Agent Prior Elicitation

A. Overview

AI-Agent Prior Elicitation (AIPE) adds exactly one new moving part to the replicated model of Section IV: a private subjective valuation $\hat{V}_i$, elicited from an AI agent, that replaces the utility trader's deterministic prior on the very first decision tick. Everything downstream of the first tick—belief blending, trust updates, utility maximisation, order matching, dividend payments, and all of the market-quality instruments—is identical to the pure Lopez-Lira run. The anchor is therefore a *minimal* perturbation of the decision model: the smallest change in behaviour that still lets the population's psychology influence its opening bids.

Two design choices make this minimal. First, the call is placed once, in parallel across all utility agents, at run start, and the result is consumed on the first decision tick and then cleared. Second, the call never asks the AI agent for an *order*; it asks only for a *number*—a valuation per share in experimental cents—which is routed through the same prior slot used by equation (10). The trader's action is still chosen by (13), using a wealth argument that the AI agent never sees. This construction isolates the psychological channel from the trading channel; it also keeps the downstream replay deterministic, because after the first tick no trader under AIPE depends on external randomness of any kind.

B. The elicitation call

Let A_U be the set of utility agents in the current population. For each $i \in A_U$ the simulator constructs a short prompt π_i that encodes three facts: the paper constants $(T, \mathbb{E}[d])$, the trader's risk preference $U_i \in \{U_L, U_N, U_A\}$ and bias mode $b_i \in \{\text{over}, \text{none}, \text{under}\}$, and the fact that the market is shared with DLM 2005 experienced counterparties. The system prompt is fixed and instructs the AI agent to return a single number with no explanation, currency symbol, or prose. Concretely, the user prompt template reads

> Market: T periods. Each share pays 0 or 20 cents with equal probability per period, expected dividend $\mathbb{E}[d]$. Risk-neutral fundamental value at run start is FV_0 cents (declines by $\mathbb{E}[d]$ per period). You, agent i, are *risk* and *bias*. You share the market with Experienced agents (Dufwenberg, Lindqvist and Moore 2005) who know the asset decays to zero and refuse to overpay in late periods. What is your private subjective valuation per share right now, in cents?

where risk $\in \{\text{risk-loving}, \text{risk-neutral}, \text{risk-averse}\}$ and bias $\in \{\text{persistently optimistic}, \text{unbiased}, \text{persistently pessimistic}\}$. All calls are fired in parallel as a single `Promise.all` batch, so the network latency is one round trip rather than $|A_U|$ round trips.

The raw reply is parsed for the first signed decimal, parsed as a float, and clamped to

$$\hat{V}_i \in [\,0.25\,\mathrm{FV}_0,\ 1.75\,\mathrm{FV}_0\,] \;=\; [25,\ 175], \tag{15}$$

so that any out-of-range reply is bounded and cannot destabilise the downstream engine. A missing reply, a parse failure, or an HTTP error silently drops the anchor for that trader; its prior then defaults to equation (10) on the first tick, and the run is otherwise unaffected. This deterministic fallback preserves the invariant that AIPE produces a simulation with or without network connectivity—a requirement for the design to be reproducible and shareable.

C. Consumption on the first decision tick

On every utility agent's first call to `updateBelief` after the simulation starts, the "psych anchor" slot is consulted in a *consume-and-clear* step:

$$V_i^{\text{prior}} \;=\; \begin{cases} \hat{V}_i \cdot (1+\varepsilon_i) & \text{if } \hat{V}_i \neq \emptyset, \\ \mathrm{FV}_t \cdot (1+b_i+\varepsilon_i) & \text{otherwise,} \end{cases} \qquad \text{then set } \hat{V}_i \leftarrow \emptyset. \tag{16}$$

The first branch overrides the deterministic prior with the AI-agent anchor and keeps the Lopez-Lira noise $\varepsilon_i \sim \text{Uniform}[-0.03, 0.03]$ so that two utility agents assigned identical anchors still draw slightly different first quotes. The second branch is the pure Lopez-Lira prior, used on every tick after the first and on every tick when no anchor is available. The "clear" step ensures that a second call to `updateBelief` on the same tick, or the next decision tick in the same run, goes through the standard path, so that the AI agent's effect is

felt at run start and then decays through the belief-blend equation (11) as normal messaging proceeds.

D. Research question and instrumentation

Let $i^\star = \arg\max_{i \in A_U} \hat{V}_i$ be the utility agent with the highest anchored valuation, and define the top holder at tick t as

$$j^\star(t) \;=\; \arg\max_{i \in A_U} \text{inventory}_i(t). \tag{17}$$

The research question is simply whether $j^\star(t) \to i^\star$ by the end of the run. The simulator reports, every tick, a boolean *match* indicator $\mathbb{1}\{j^\star(t) = i^\star\}$ and a *normalised gap*

$$g(t) \;=\; \frac{\hat{V}_{i^\star} - \hat{V}_{j^\star(t)}}{\hat{V}_{i^\star}} \;\in\; [0, 1], \tag{18}$$

together with the identities of the two agents and their valuations. The gap collapses to zero when the asset has reached the psychologically-maximal holder; it stays near one when the asset has migrated to an agent whose anchor is much lower than the maximum. Figure placeholders in the `figures/` directory will host time-series plots of $g(t)$ and the match indicator for each of the three paradigms.

The match indicator is related, but not identical, to the allocative efficiency statistic

$$\text{AE} \;=\; \frac{\sum_i \hat{V}_i \cdot \text{inventory}_i(T)}{\max_i \hat{V}_i \cdot Q}, \tag{19}$$

in which the numerator is the realised cross-agent utility weighted by each agent's anchored valuation and the denominator is the best-possible allocation in which every share is held by the maximum-valuation agent. When $\text{AE} = 1$ the asset has migrated to $i^\star$ and the match indicator is true; but AE can be close to one even when the match indicator is false, for example when the top holder is different from $i^\star$ but has a nearly-maximal anchor. Reporting both statistics together is therefore informative: the match indicator captures the identity of the eventual owner, while AE captures the continuous loss relative to the psychological optimum.

E. Determinism, cost, and caching

A single AIPE run consumes $|A_U|$ chat completions as one parallel batch and runs closed-form thereafter. With $|A_U| \leq 6$ in the default preset, network cost is a handful of round trips on the first frame and zero afterwards; no caching or retry layer is needed because a missing anchor silently degrades to the deterministic prior. The seeded PRNG makes identical anchor vectors reproduce identical trajectories, so within-model replay is trivial and across-model comparisons reduce to toggling the paradigm button.

VI. Preliminary Results

Because the simulator is designed for interactive replication rather than a single canonical experiment, this section reports preliminary findings from three paradigm presets rather than a calibrated empirical test. The *strict-DLM* paradigm runs the exact protocol of Dufwenberg et al. (2005): $N = 6$ traders, two discrete endowment types, four consecutive rounds, the round-4 replacement treatment, and the ten-session batch. The *Lopez-Lira* and *AIPE* paradigms continue to use the relaxed population size and sampler, since they are extensions rather than replications. Unless otherwise noted the seeded PRNG is rolled fresh at the start of every run, and the AIPE anchor is drawn from the default model with the paper's system and user prompts verbatim. The market-quality statistics reported below are pooled across all four rounds of a session; the per-round breakdown that drives the experience-kills-bubbles finding is described in the baseline DLM subsection.

A. Baseline strict-DLM: experience kills the bubble

Under strict-DLM mode the six traders are all instantiated as `DLMTrader` instances with `roundsPlayed`=0, so every subject in round 1 operates on the inexperienced branch of equation (4). The round-1 tape therefore runs into the bubble region: amplitude A typically exceeds 0.35, ND exceeds 0.20, and Haessel's R^2 is near zero because transaction prices overshoot the staircase in periods 2–5 and crash in periods 8–10. At the end of round 1 the engine increments every survivor's `roundsPlayed` to 1 and the traders flip to the experienced branch for round 2. Amplitude falls, ND contracts toward 0.10, and R^2 rises to above 0.50, even though the

population is identical, the seed is the same, and no rule has changed. By round 3 the bubble is essentially absent, and round 4 — under either treatment — matches the staircase to within a few experimental cents for the veteran subset. The simulator reproduces the qualitative DLM 2005 finding without pre-assigning experience at any point.

The round-4 replacement treatment manipulates the veteran/fresh mix: T2 leaves four veterans and two fresh subjects, T4 leaves two veterans and four fresh subjects. In preliminary runs, T4 widens the round-4 price band compared to T2 because the four fresh traders generate enough momentum-chasing on the inexperienced branch to drag the book above the veterans' liquidation threshold for a period or two before they dump; the size of the effect is the research question the ten-session batch is built to measure. Batch summaries print the mean normalised deviation and the mean per-subject payoff (§H, equation (5)) for each treatment so that side-by-side comparisons are possible in a single click.

B. Lopez-Lira 2025: the strategy cube shapes the bubble

Under the Lopez-Lira preset (six utility agents, no experienced), the price dynamic depends sensitively on the population's *strategy cube*: bias mode $\times$ belief mode $\times$ risk preference. When $\alpha_L > 0.6$, the convex $U_L(w)$ drives aggressive ask-crossing in early periods with amplitude $A > 0.40$, positive residuals through periods 1–7, and a terminal collapse—the canonical SSW bubble-and-crash shape with Haessel R^2 near zero. When $\alpha_A > 0.6$, concave $U_A(w)$ pulls residuals negative and raises ND into the 0.15–0.25 range, reflecting under-pricing rather than bubble formation. Toggling the deceptive messaging mode produces the largest perturbation: deceptive broadcasters drag the trust EMA down once the period-end closeness update of (12) runs, and the mean-lie-magnitude statistic is tightly correlated with ND, consistent with the closed-loop mechanism in Lopez-Lira (2025).

C. AIPE: does the asset migrate to the highest anchor?

The novel regime is the AIPE merged market. The Utility agents sit alongside the fixed $F/T/R$ background; no agent is pre-labeled as experienced. A live AI-agent anchor vector $(\hat{V}_i)_{i \in A_U}$ is drawn at run start, and three preliminary observations are clear from a handful of runs.

Anchor dispersion.. The returned anchors are heterogeneous across risk and bias archetypes in the expected direction. Risk-loving, persistently-optimistic agents receive anchors near the upper clamp $1.75\,\mathrm{FV}_0 = 175$; risk-averse, persistently-pessimistic agents receive anchors near the lower clamp $0.25\,\mathrm{FV}_0 = 25$; risk-neutral, unbiased agents cluster near $\mathrm{FV}_0 = 100$. The simulator reports the full anchor vector in its subjective-valuation panel, so the reader can verify that the model's psychology maps into $\hat{V}$ in the direction the prompt implies.

First-tick quoting.. Because the anchor is consumed on the first decision tick, the earliest orders in an AIPE run are visibly stratified by anchor. Risk-loving agents with $\hat{V} \approx 175$ post aggressive bids near $\mathrm{FV}_1 = 100$ and cross the ask when possible; risk-averse agents with $\hat{V} \approx 25$ post asks well below FV_1. The effect lasts roughly one period before the Lopez-Lira belief-blending channel pulls everyone back toward FV_t, but it produces a clear opening asymmetry absent from a pure Lopez-Lira run with the same seed.

The match indicator.. Across runs, the boolean match indicator $\mathbb{1}\{j^\star(t) = i^\star\}$—whether the agent currently holding the most shares is the agent with the highest anchor—is not monotone in t. It typically starts near the seed endowment distribution (which is independent of $\hat{V}$), rises during periods 1–3 as the aggressive high-anchor agents accumulate inventory, and then drifts in the terminal region as the fundamentalist background liquidates its shares. In runs where the highest-anchor agent is also the fundamentalist bubble's counterparty, the indicator ends at one; in runs where the background liquidates into an agent of middling anchor, the indicator ends at zero and the corresponding normalised gap $g(T)$ is typically in the 0.10–0.30 range. The allocative efficiency statistic AE moves in the same direction but stays strictly positive in every run because the denominator in (19) is $\max_i \hat{V}_i \cdot Q$ and the numerator always contains a nontrivial contribution from the top holder.

D. Market-quality statistics at a glance

Table 1 summarises the canonical bubble statistics under the three paradigms in qualitative—not yet calibrated—form.

Table 1: Market-quality statistics, qualitative ranges

	DLM r. 1	DLM r. 4	Lopez-Lira 2025	AIPE
Haessel R^2	≈ 0	> 0.80	0.30–0.80	0.40–0.85
Normalised dev. ND	> 0.20	< 0.10	0.10–0.30	0.10–0.25
Amplitude A	> 0.35	< 0.15	0.20–0.50	0.20–0.45
Turnover TO	0.6–1.1	0.3–0.6	0.6–1.2	0.5–1.0
Allocative eff. AE	0.6–0.8	≈ 1	0.6–0.9	0.7–0.95
Match $\mathbb{1}\{j^\star(T) = i^\star\}$	n/a	n/a	n/a	50–70% runs

The two strict-DLM columns ($r = 1$ and $r = 4$) together recover the DLM finding: a large bubble in round 1 with the same population that produces a tight staircase in round 4, separated only by the three intervening rounds of procedural experience. The Lopez-Lira and AIPE columns are pooled across the full session rather than split by round because their relaxed populations do not have an endogenous round-1 to round-4 sharpening channel — the agents keep the same trust/belief state across rounds but none of them gate behaviour on `roundsPlayed`. AIPE sits between strict-DLM round 4 and Lopez-Lira because the fundamentalist background still suppresses amplitude and improves R^2, while the utility block introduces enough psychology to produce nontrivial variation in the match indicator. Whether the match indicator exceeds a naive baseline in a properly replicated experiment — and whether the anchor adds predictive power beyond the deterministic prior — is the question the testbed is built to answer.

VII. Conclusion

This paper documents a browser-based replication of the experimental asset market of Dufwenberg et al. (2005), extended with the expected-utility messaging population of Lopez-Lira (2025), and introduces *AI-Agent Prior Elicitation* (AIPE), a protocol in which a single AI-agent call seeds each utility trader's private subjective valuation at run start. The extension is deliberately minimal: one number per agent, clamped to $[0.25\,\mathrm{FV}_0,\ 1.75\,\mathrm{FV}_0]$, written to the `psychAnchor` slot and consumed on the first decision tick of the standard Lopez-Lira prior. Everything downstream of the first tick—the belief blend, the trust EMA, the expected-utility action choice, the order matching, the dividend crediting, and the market-quality instruments—is unchanged from the replicated baseline, and the simulator's seeded mulberry32 generator ensures that every run can be replayed identically with or without network access.

The contribution is threefold. First, the testbed constitutes a clean vanilla-JavaScript implementation of two otherwise unrelated experimental designs, specified in enough detail (Sections III–IV) that a reader can reconstruct the model from the paper alone. Second, AIPE introduces a narrowly-scoped use of an AI agent as a *psychological prior elicitor* rather than as a trader: the AI agent is never asked to choose an action, and its output is routed through the same prior slot used by the deterministic fallback, so that any behavioural effect is attributable to the starting valuation rather than to an expansion of the action space. Third, the research question—does the asset end up in the hands of the agent with the highest anchored valuation—is instrumented with a per-tick match indicator and a normalised gap, and reported alongside the classical bubble statistics (Haessel R^2, normalised deviation, amplitude, turnover, allocative efficiency). The instrumentation is designed so that any replication of AIPE can report the same panel without additional code.

Three caveats shape the preliminary findings. First, the ranges in Table 1 are illustrative, not calibrated: a full empirical test requires many replications per cell, a pooled treatment of the network variance of the anchor, and a formal baseline against a deterministic prior with matched mean and variance. Second, the anchor is consumed exactly once, which bounds the downstream signal at whatever the first-tick prior contributes to the five-action expected-utility score; refreshing $\hat{V}_i$ each period would be a natural extension but is in tension with

the reproducibility invariant. Third, generalising beyond the DLM 2005 dividend distribution and horizon would require re-deriving the market-quality instruments and re-calibrating the DLM trader's experienced-branch horizon discount $\varphi(h)$ described in §D.

The testbed is free of build steps and of third-party dependencies, and ships with the architecture diagram, glossary, and slide deck used in this paper. Its "Experiment settings" panel exposes every numerical constant that shapes a run—the population mix, the risk-preference share, the passive fill probability, the trust learning rate, the bias amplitude, and the valuation noise—and the seeded PRNG makes every edit to a constant observable as a difference against a matched baseline. AIPE is the first extension of this infrastructure that asks a question about AI agents, and it opens several follow-ups: whether the anchor's effect on allocative efficiency scales with the cross-agent variance of $\hat{V}$; whether different AI agents produce systematically different anchor vectors for the same prompt; and whether adding deceptive messaging amplifies or dampens the first-tick stratification that the anchor produces.

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